

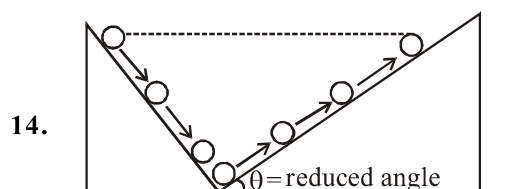
Hints & Solutions

6. Max. friction depends on mass of the body too.
 8. Slope of K.E – x graph is

$$\frac{d\left(\frac{1}{2}mv^2\right)}{dx} = \frac{1}{2}m \frac{2v dv}{dx}$$

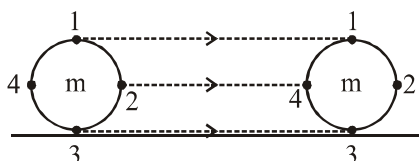
$$= \text{const} \times \frac{v dv}{dx} = \text{const} \times \text{acc}^n$$

13. $\Delta P = \int F dt$

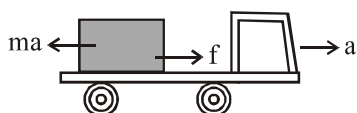


Final height is same as initial height.

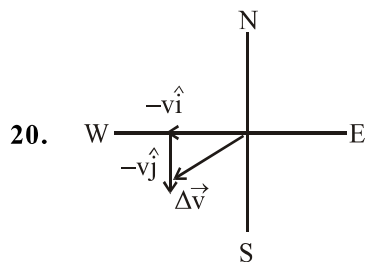
15. Uniform translatory motion



16. Without force object having acceleration implies that frame of reference is having acceleration.
 18. Pseudo force on block acts in backward direction. There force direction of friction force on the box in forward direction.



19. Friction force between the tyre & road causes the acceleration of car on horizontal road.

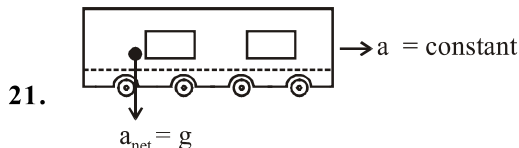


$$\Delta \vec{v} = \vec{v}_f - \vec{v}_i$$

$$\Delta \vec{v} = -v_i \hat{i} - v_j \hat{j}$$

$$\Delta \vec{v} = \sqrt{2}v, \text{ south-west}$$

Change in velocity takes place in direction of $\vec{F}_{\text{net}}$ [Frictional force along south-west]



Force on the body at a given time is determined by the situation at the location of the body at that time. Force is not carried by the body from its earlier history of motion.

22. According to Newton's 3rd law

System



$$\vec{F}_{AB} = -\vec{F}_{BA}$$

$$\vec{F}_{AB} + \vec{F}_{BA} = \vec{0}$$

23. $mv_1 > mv_2$
 momentum of 1st is greater than that of 2nd
 27. Protons have mass and charge. They are also nucleons.
 28. Body always moves in the direction of velocity, if velocity is zero then moves along net force on body.
 29. If the spring is pushed then it pushes the body; if the spring is pulled then it pulls the body.
 30. Equilibrium implies $\vec{F}_{\text{Net}} = 0$, thus $\vec{a} = 0$ or $\vec{v} = \text{const}$, velocity must be const and in magnitude be constant in magnitude as well as in direction. The const velocity may be zero.
 33. $\mu_s > \mu_k$; $0 \leq f_s \leq \mu N$ and $f_k = \mu_k N$
 34. Smaller is the value of friction, lesser is the horizontal force required to displace the block.
 In case (1) $f_{\text{max}} = \mu N = \mu mg$
 In case (2) $f_{\text{max}} = \mu(mg - F \sin \theta)$
 In case (3) $f_{\text{max}} = \mu(mg + F \sin \theta)$
 In case (4) force required is maximum
 35. According to Newton's 1st law.
 36. If $\vec{F}_{\text{Net}} = 0$ on a body then $\vec{a} = 0$
 37. Force does not depend upon frame of reference.
 38. According to Newton's 1st Law.

39. Gas will come out with sufficient speed in forward direction so reaction of this forward force will change the reading of the spring balance.
40. As by an internal force momentum of the system can not be changed.
41. In case A, it is elongation and in case D, it is compression.
43. Lift is descending with retardation. Relative to lift force in downward direction is $m(g+a)$ in place of mg . Therefore on smooth plane acceleration will be $(g+a) \sin\theta$ not $g\sin\theta$.
45. With no external force, object can be stationary or move on line with uniform speed. In this situation particle can not move on circle because to move on circle or curve external force is necessary.